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Review

The role and value of self-employed nurses in the allocation of nursing human resources: A scoping review

Nianqi Cui [a,1], Wei Yang [b,1], Yulin Lu [a], Jingfen Jin [c,*]

[a] School of Nursing, Kunming Medical University, Kunming, China
[b] Faculty of Medicine, Yunnan College of Business Management, Kunming, China
[c] Nursing Department, The Second Affiliated Hospital of Zhejiang University School of Medicine, Hangzhou, China

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ABSTRACT

Objective: This study aimed to conduct a scoping review to determine the role and value of self-employed nurses in the allocation of nursing human resources.
Methods: Following the scoping review methodological framework proposed by Arksey and O'Malley, China National Knowledge Infrastructure (CNKI), WanFang, CINAHL, PubMed, Web of Science, and Scopus databases were searched from inception to February 2025 on self-employed nurses in the nursing field. Thematic content analysis was used to collate, summarize, and narratively report themes and subthemes on nursing data practices.
Results: A total of 60 articles were ultimately included. Four themes emerged from the scoping review. Among them, 14 articles focused on the qualification certification and competency requirements of self-employed nurses, 24 articles delved into the roles and responsibilities, 24 articles focused on the value, and the remaining four articles focused on the registration and legalization. Self-employed nurses must have a bachelor's degree in nursing and pass relevant exams, while specialist nurses need even more skills. They offer personalized care through clinics or home visits, providing a flexible and autonomous career path, reducing healthcare costs, and encouraging interdisciplinary collaboration.
Conclusions: This study highlights the various roles and values of self-employed nurses. However, self-employed nurses face challenges, including nonuniform regulations, practice risks, a lack of medical insurance policies, shortages in remote areas, and inadequate educational support. Key strategies for enabling effective contributions to the healthcare system include the standardization of entry requirements, the strengthening of certification processes, the enhancement of safety measures, the development of medical insurance policies, the attraction of talent, and the reinforcement of educational frameworks.
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What is known?

- In the global shortage of nurses and an aging population, self-employed nurses play an integral role.
- The lack of clarity regarding information about self-employment hinders nurses from choosing to be self-employed.

What is new?

- Self-employed nurses must complete a bachelor's degree in nursing and pass relevant examinations, with more stringent requirements for specialist nurses. They can provide services to patients in clinics or through home visits. This practice offers a new career development pathway for health care workers, reduces health care system costs, and promotes interdisciplinary collaboration.
- Self-employed nurses currently face multiple challenges, including nonuniform practice regulations, risks, a lack of medical insurance policies, shortages of personnel in remote areas, and inadequate educational support.

* Corresponding author.
E-mail address: zrjzkhl@zju.edu.cn (J. Jin).
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[1] Nianqi Cui and Wei Yang contributed equally to this study.

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- To boost self-employed nurses, we need comprehensive policy changes: implement a certification system with flexible qualifications, improve risk prevention through targeted training and collaboration, regulate service pricing with fee caps and incentives, and focus on nursing entrepreneurship education with specialized courses and practical skills.

1. Introduction

Self-employed nurses are registered nurses who operate independently, either as independent practitioners, partners, or employers, rather than as employees of health care institutions [1]. They are directly accountable to clients for whom they provide nursing services [1,2]. The concept of self-employed nursing emerged in the early 1970s in the United States. It was driven by nurses' dissatisfaction with the traditional health care system and their desire for greater professional autonomy. This led to the development of an independent nursing practice model in which nurses provided services individually or as part of a team. As the demand for autonomy in professional control, time, and finance increases, more nurses seek self-employment opportunities to return to clinical care through independent practice and provide high-quality nursing care while earning financial rewards commensurate with their value. Consequently, in 1979, the Independent Nurses Association (INA) was established, allowing members to be self-employed [3].

With the establishment of official organizations, the number of self-employed nurses has also increased in different countries. Self-employed nurses first emerged in the United States, and by 1980, the number of self-employed nurses in the United States had increased significantly, with nearly 11,000 nurses being self-employed [4]. At the beginning of the 21st century, the International Council of Nurses (ICN) data estimated that 0.5 %–1 % of registered nurses worldwide were self-employed [5]. In the United States, self-employed nurses account for approximately 0.18 % of nurses [5,6], whereas in the United Kingdom (UK), the proportion is as high as 18 % [7]. In one Canadian province, among 29,000 registered nurses, 241 were self-employed [8].

Self-employed nurses assume distinct roles within the health care system and contribute diverse values. Roles refer to individuals' social positions throughout their life cycle [9]. Self-employed nurses undertake their roles in two main ways. First, they can open clinics directly. Self-employed nurses can work independently or collaborate with other nurses to form a nursing team, choosing to provide nursing care either through home visits or within the clinic [10]. In addition, they can work with doctors to manage patients jointly [11]. Second, self-employed nurses can work with medical institutions or national health systems. For example, in the UK National Health Service (NHS), self-employed nurses sign contracts with patients to provide free nursing care, with the clinic being paid by the central government [12,13].

While performing their roles, self-employed nurses add significant social value. Value refers to the importance, value, or utility of something, a widely recognized concept that measures peoples' or things' significance and positive attributes [14]. Self-employed nurses proactively provide essential home nursing and chronic disease management, offering professional long-term care (e.g., wound management, injections) for homebound, disabled, and elderly patients, meeting urgent needs for home-based care [15]. Regular follow-ups allow timely adjustments to care plans, reducing complications and readmissions [8]. Within the medical system, they contribute through specialized/policy consultation and community health services, diversifying nursing roles [16]. In remote areas, they establish clinics providing accessible, affordable care and preventive education, improving community health and economic prospects [8,17]. Amid global aging and nursing shortages, they are vital in addressing chronic, rehabilitative, and long-term care needs.

Currently, research on self-employed nurses has made significant progress in some aspects. Studies identify entrepreneurial motives: internal factors include pursuing nursing autonomy, practicing values-aligned nursing concepts, and career flexibility; external factors relate to healthcare resource shortages [5,9,18–25]. Other research examines ethical conflicts, balancing profit-seeking goals with nursing ethics and the impact of industry stereotypes on professional identity [26–28]. Consumer perception studies have investigated public and patient attitudes towards self-employed nurse services [29]. Research has analyzed the relationship between self-employed nurses' innovative traits and entrepreneurial behaviors, exploring the effect of innovation ability on practical success [27,30].

While research on self-employed nurses exists, their roles and values remain understudied. First, self-employed nurses' presence is largely confined to certain developed countries with varying registration, certification, and competency standards [17,31–34], hindering establishment elsewhere due to a lack of uniform frameworks. Second, administrative hurdles, such as delayed licensure from authorities and insufficient managerial training, impede clinic development [24,32,35]. Furthermore, significant challenges exist in obtaining medical insurance reimbursement and accessing occupational benefits [36,37]. The arduous reimbursement process, differing from traditional employment models, creates financial burdens, restricts resource access, and negatively impacts well-being [9,36]. In addition, the specific service forms provided and potential categorization models for future reference remain unclear. Finally, the value of self-employed nurses has not been systematically summarized. Existing fragmented research cannot provide a robust basis for policy recommendations, hindering the acquisition of essential policy support.

Consequently, this study aimed to conduct a scoping review to analyze the qualification certification and competence requirements of self-employed nurses, to clarify the role of self-employed nurses within the health care system, and to explore the value of their development for the nurses themselves and their patients. It bridges knowledge gaps and offers references for their integration into global health care, promoting higher-quality global nursing development in diverse service models.

2. Methods

2.1. Study design

This research utilized the Arksey and O'Malley framework to conduct a scoping review [38], which involved: 1) identifying the research questions; 2) identifying relevant studies; 3) study selection; 4) charting the data and 5) collating, summarizing, and reporting the results. This study adhered to the PRISMA extension for scoping reviews (PRISMA-ScR) [39]. This approach has been widely utilized in nursing research to explore and analyze complex questions because it facilitates a comprehensive mapping of literature and supports the development of conceptual frameworks [40]. By adopting this methodology, the review ensures a systematic and transparent process.

2.2. Study process

2.2.1. Identifying the research questions

This review focuses on four primary questions: a) What

qualification, certification, and competence requirements do self-employed nurses possess? b) What are the roles and responsibilities of self-employed nurses? c) What is the value of self-employed nurses? d) In which countries or regions are self-employed nurses legally accepted?

2.2.2. Identifying relevant studies

A comprehensive search strategy (Appendix A) was developed and applied to six major databases from inception to February 2025: China National Knowledge Infrastructure (CNKI), Wanfang, CINAHL, PubMed, Web of Science, and Scopus. The search strategy incorporated free-text terms and Medical Subject Headings (MeSH) combined with Boolean operators to ensure a thorough retrieval of relevant articles. In addition, grey literature was identified through Google Scholar, and both backward and forward citation tracking were conducted for all included studies.

To determine the inclusion and exclusion criteria, the population-concept-context (PCC) framework was employed [41]. The population refers to self-employed nurses. The concept encompasses the role and value of self-employed nurses within the nursing domain, including their qualification certifications and competency requirements, roles, responsibilities, and value. The context is nursing practice. The inclusion criteria were as follows: a) articles involving self-employed nurses or related independent nursing practitioners; b) articles exploring the role and value of self-employed nurses; and c) articles that included but were not limited to qualitative research, quantitative research, mixed-methods research. The exclusion criteria were as follows: a) abstracts and conference papers; b) articles mentioning the topic only in the title, abstract, or keywords without substantive discussion in the main text; c) articles with unavailable full texts; and d) duplicate publications.

2.2.3. Study selection

Studies identified in the initial phase were screened by title and abstract; uncertain eligibility prompted full-text review for final selection. Records failing to meet the inclusion criteria were excluded from this scoping review. Two researchers performed all selection steps; disagreements on eligibility were resolved by consensus among the authors.

2.2.4. Charting the data

A data-extraction form developed in Microsoft Word captured information from the selected studies. Data points from all the included articles were gathered: author and publication year, aim or objective, methods, relevant findings or results, reported nursing care, and usefulness to the review.

2.2.5. Collating and summarizing the results

The research employed thematic content analysis, with the themes and subthemes related to nursing data practices being collated, summarized, and presented in a narrative report.

3. Results

A total of 10,779 potentially relevant articles were identified from the databases and imported into EndNote X9. A total of 60 articles were ultimately included [1,2,4,8,10–13,15,16,18,20–22,24–26,28,29,36,37,42–80] (Appendix B). Among them, 14 articles focused on the qualification certification and competency requirements of self-employed nurses [4,12,15,18,24,37,42–49], 24 articles delved into the roles and responsibilities of self-employed nurses [1,2,10,12,13,21,50–67], 24 articles concentrated on the value of self-employed nurses [8,11,13,16,20,22,25,26,28,29,36,51,56,68–78], and the remaining four articles focused on the registration and legalization [12,15,79,80]. The detailed information of all the included studies is shown in Appendix C.

3.1. Qualification certification and competency requirements of self-employed nurses

3.1.1. Career advancement

In some countries, such as China, the titles of nursing professionals are categorized into three tiers: primary (nurse, nurse practitioner), intermediate (nurses in charge), and senior (associate chief nurse, chief nurse) [34]. Newly recruited nurses can obtain primary titles after passing the practice qualification exam. Promotion to nurse practitioner requires specific educational and work experience criteria: 5 years for secondary specialized school graduates, 3 years for junior college graduates, and 1 year for undergraduates or above. The intermediate title necessitates meeting educational and seniority requirements: 7 years as a nurse practitioner for secondary specialized school graduates, 6 years for junior college graduates, 4 years for undergraduates, and 2 years for master's degree holders. Candidates must also pass the national chief nurse qualification exam. The senior titles include associate chief nurse and chief nurse. Associate chief nurse demands 5 years in charge and documented leadership in nursing quality improvement projects. Promotion to chief nurse typically requires 5 years as associate chief nurse and essential management experience.

However, self-employed nurses have not yet established a clear career development path with a singular identity. Self-employed nurses advance professionally by enhancing competencies through the continuing competence program (CCP), such as by completing quality improvement (QI) learning plans and specialized training (such as nursing foot care and aesthetics) [12,15]. They broaden their practice areas to include administration, such as health care staffing agencies, or consultation, like health system navigation, to ensure their services comply with legal scopes and evidence-based standards. Creating networks for interprofessional collaboration for referrals and partnerships, along with participating in risk management and policy development, such as updating practice guidelines, enhances professional credibility [15]. Some may become industry experts through focused practice areas or contribute to regulatory committees, influencing nursing standards.

3.1.2. Qualification certification

Like other nursing professionals, self-employed nurses begin their careers by completing an Associate Degree in Nursing (ADN) or a Bachelor of Science in Nursing (BSN) from an accredited nursing school and passing the National Council Licensure Examination for Registered Nurses (NCLEX-RN). While an advanced degree is not typically required for self-employed nurses to start their own business, some nurses may consider a dual master's degree program in nursing and business administration [a dual Master of Science in Nursing (MSN) and Master of Business Administration (MBA)] to understand the industry better and identify business opportunities within it [42].

Self-employed nurses must hold a professional registered nurse license and practice legally in their respective regions, ensuring the services comply with local nursing practice acts [43]. When engaging in specialized nursing fields, additional requirements include a) a master's degree in the relevant nursing field, emphasizing both theoretical knowledge and clinical experience; b) regular supervision by qualified professionals; c) at least two years of diverse clinical experience post master's degree, including individual, group, or family therapy and crisis intervention; and d) in some regions, a certification as a clinical nursing specialist in the

field or evidence of meeting the certification standards is required, which indicates the professional competence of the practitioner in that area [44].

3.1.3. Competency requirements

Numerous scholars have indicated that self-employed nurses are required to possess comprehensive competencies [4,12,18,24,36,45–48]: a) solid professional competencies to provide high-quality nursing care for individuals, families, groups, and communities, including knowledge, skills, and judgment in the nursing profession; b) advocacy competencies to provide safe, competent, compassionate, and ethical care for patients; and c) the ability to communicate effectively with patients, family members, and institutional staff and to establish interpersonal networks. Listening to and understanding patients' experiences, preferences, and health goals, as well as documenting and reporting clearly and accurately, are vital; d) business skills, including risk reduction and management, marketing, etc., enable self-employed nurses to find their niche in the competitive health care market and enhance their services' competitiveness and economic viability. Additionally, Rew [49] introduced the nursing AFIRM model, which is grounded in role theory, to assist clinical nursing specialists in planning their self-employed practices. The acronym AFIRM represents nurses' steps to develop their roles in a self-employed context: availability, formulation, factual information, referrals, and monitoring. Each step includes specific questions to address during planning to increase nurses' proficiency in this role.

3.2. The role and responsibilities of self-employed nurses

The role of self-employed nurses is primarily providing health care, and the working methods of self-employed nurses can be divided into two categories: direct and indirect [13]. Direct methods include nurses who work for themselves, providing care to patients and directly charging them fees [1,2,50]. Some studies [51–54] indicate that self-employed nurses can operate their clinics or work together to offer treatment services. They can offer services such as diabetes education, management, and stoma care at patients' homes or clinics in areas not within hospitals. In ophthalmology clinics led by ophthalmic nurses, self-employed nurses can provide nursing services for patients with cataracts and glaucoma, reducing patient waiting times and enhancing treatment safety [55]. Research by Currie et al. [56] has indicated that self-employed nurses spend an average of 75 % of their weekly time on direct patient care. They conduct independent diagnoses and prescribe medications, following up on patients' health conditions. Moreover, most self-employed nurses collaborate with public health professionals to refer patients to other health professionals to reduce duplication and fragmentation of care [10,53,56]. Indirect methods involve working with national health systems, such as the UK's NHS [57–60], where self-employed nurses collaborate with the NHS to offer individual health care services to residents, who directly engage with the clinic and benefit from free NHS services [12,13]. This means that the central government purchases the service, and the NHS, which the central government finances, pays the clinic according to the number of individuals registered. As a result, self-employed nurses are incentivized to deliver a more extensive array of health care services, keep costs in check, and receive higher pay than NHS staff with fixed incomes [12]. This can significantly broaden the range of primary health care services accessible to the public and enhance health care access [61].

In addition to health care, self-employed nurses engage in nursing consultation, education, and research [50,62–64]. Educational services include communication skills, community education, and complementary therapies [62]. Self-employed nurses typically spread health knowledge to nurses, other health professionals, health care institution staff, and the public through courses or conferences [56,65]. As consultants [66], self-employed nurses provide clients such as hospitals, universities, and nursing homes with consulting services related to their expertise to address problems [21,52,67]. A small number also participate in research, offering advice and assisting with funding applications [62].

3.3. Value of self-employed nurses

The value of self-employed nurses to patients is demonstrated through the following. 1) Home visits: self-employed nurses provide long-term, professional home-based medical and nursing care for elderly individuals with disabilities or chronic illnesses. These patients often face difficulties in accessing hospitals because of mobility issues and a lack of companions. By offering home visits, self-employed nurses alleviate the significant burden of caregiving on society and families, thereby improving the quality of life for these patients [68]. Additionally, specialized maternity and infant care nurses can provide home visits to meet the specific needs of postpartum women, better meeting their individualized home care requirements after discharge from hospitals. This enhances the health literacy of both mothers and their families, potentially reducing the incidence of postpartum complications [69]. 2) Personalized nursing care: self-employed nurses are able to offer highly individualized and comprehensive nursing services on the basis of the specific needs of patients. Such high-quality care caters to the demands of individuals with better economic conditions who seek superior nursing services [8,20,29]. For example, self-employed nurses can provide 24-h specialized care for children requiring special medical attention, including the management of ventilators, tracheostomy care, and nutritional support [70]. 3) Enhancing access to nursing care: self-employed nurses can establish clinics in areas where traditional health care services are insufficient and can thus provide care to low-income individuals, rural residents, and those not covered by the national health care system [26,28,35,71,72].

Self-employed nurses provide the following values to health care professionals. 1) Offering new career pathways: self-employed nurses provide a new direction and possibilities for career development for other nurses, nursing students, and midwifery students [73]. They demonstrate how to achieve personal value and career satisfaction beyond traditional nursing roles through innovative practices. These successful examples inspire more health care professionals to consider entrepreneurship and explore new career opportunities [8,25]. 2) Providing career flexibility and autonomy: unlike traditional nurses who are strictly bound by schedules and confined to working within hospitals [16,26], self-employed nurses can arrange their working hours and locations flexibly through independent management, thus achieving a better work–life balance [13,22,74,75].

Self-employed nurses provide value to the health care system in two main ways. 1) Reducing health care costs: firstly, the global health care system is facing the significant challenge of nurse shortages. To address this issue, many hospitals opt to hire self-employed nurses or purchase the services of self-employed nurses as short-term solutions. These nurses typically do not receive social security benefits, thereby reducing the health care costs for hospitals [76,77]. Second, self-employed nurses can manage 80 %–90 % of health issues that primary care physicians can handle without the need for referrals or consultations. Through community nursing and telemedicine, patients' hospital

stays can be reduced, and medical resources can be conserved [22,56,71]. Additionally, the expanded role of self-employed nurses in primary care not only provides medical service of a quality comparable to that of physicians but also has a lower training cost than do general practitioners or primary care physicians. This can significantly save resources for the health care system [78]. 2) Promotion of interdisciplinary collaboration: self-employed nurses collaborate with other professionals in practice, such as physicians, physical therapists, and dietitians, providing comprehensive medical services. This collaborative model helps to enhance the overall quality of medical services [11,20,51].

3.4. Registration and legalization

There is a lack of clear and unified registration standards for self-employed nurses. Typically, provincial regulatory bodies handle the annual registration update process, which varies across regions. In Canada, a relatively developed and representative country, licensed practical nurses (LPNs) in Manitoba who practice self-employment must maintain active registration with the College of Licensed Practical Nurses of Manitoba (CLPNM), completing at least 250 practice hours annually in their self-employed capacity. They must confirm their business name and self-employed status on the annual registration renewal form, submit the self-employed practice business name registration form and questionnaire within 30 days of establishing practice, and fulfill the CCP requirements, including a QI learning plan relevant to their practice. If selected for a CCP audit, they must provide nursing care plan templates and policy manual excerpts for review [79,80].

Saskatchewan's self-employed Registered Nurses (RNs) and Nurse Practitioners (NPs) must apply for Recognition of Practice (ROP) with the College of Registered Nurses of Saskatchewan (CRNS), providing detailed written descriptions of their services and demonstrating alignment with practice standards and entry-level competencies. They are required to complete 1,125 practice hours over five years (900–1,000 h for NPs) and participate in the CCP, including peer feedback. The ROP process ensures that services align with the legislative scope, and failure to obtain approval may affect liability coverage and practice hour validation [12,15]. In Alberta, while the College of Registered Nurses of Alberta (CRNA) mandates 1,125 practice hours over five years, the process for ROP remains unclear, with conflicting information on application requirements [15]. Ontario's College of Nurses does not specify minimum practice hours but requires a Declaration of Practice confirming nursing engagement within the past three years [15].

4. Discussion

4.1. Qualification certification and competency requirements of self-employed nurses

Self-employed nurses must hold a professional RN license and practice legally within their jurisdiction. Those possessing a master's degree and at least two years of diverse clinical experience are qualified to practice in specialized nursing fields. Self-employed nurses benefit from shorter training periods, less stringent practice standards, and quality-assured frameworks. This constitutes their core advantage in enabling safe, effective, and sustainable practice outside traditional hospital systems. However, in countries like China, "online nurses" function similarly to self-employed nurses, using an "online application, offline service" model to care for discharged patients, elderly individuals with disabilities, and others with mobility-limiting illnesses [81]. "Online nurses" must be registered with medical institutions and possess at least five years of clinical nursing experience and a professional title of nurse practitioner or above [81–84]. To work in a specialized field, they must have a qualification certificate from provincial or municipal specialist nurse training programs and complete training with assessments by local nursing societies or affiliated medical institutions [83,85,86]. Therefore, one possible suggestion is to allow self-employed nurses specializing in particular fields to reduce the required clinical work experience from five to two years after completing a three-year graduate program. This approach may shorten the training period and potentially facilitate the expansion of the self-employed nurse workforce.

In China, only 21.5 % of specialist nurses work full-time in their field, while most with specialist certifications do not fully apply their skills in practice. This is due to limited job opportunities, unclear roles, and insufficient institutional support, resulting in little distinction between the duties of specialist and general nurses, and a focus on training over practical use [30,87,88]. Therefore, if these specialist nurses were to transition to self-employment and apply their specialized knowledge and skills to broader and more needy patient populations, it would better leverage their professional strengths and provide them with broader career development opportunities. Moreover, in the context of a challenging employment situation, vacated positions could offer new graduates the opportunity to enter clinical practice and accumulate experience, thereby creating a virtuous cycle. This would help alleviate the shortage of nursing human resources and optimize the allocation of medical and nursing resources, ultimately increasing the mobility of nursing staff and maximizing their potential.

4.2. The role and responsibilities of self-employed nurses

This study indicates that self-employed nurses are capable of independent practice and providing home visits to patients. This not only enhances the accessibility and continuity of nursing care but also promotes patient-centered care, while simultaneously mitigating strain and cost burdens on the healthcare system. Studies on "online nurses", like self-employed nurses, highlight the need to enhance their risk prevention and emergency training. However, risk issues are important and inescapable barriers on the path of self-employed nurses. First, home visits differ from nursing services in medical institutions. Self-employed nurses encounter many risks during service delivery, such as changes in patient conditions and shortages of medical supplies. At the same time, risk prevention and emergency training for these nurses are lacking [89,90]. Therefore, self-employed nurses should be trained to prevent and assess risks during home visits, identify and manage common emergencies. Moreover, the government should introduce policies to facilitate information sharing and improve the management system to provide medical resource support [91].

Second, female nurses who primarily provide home visits often exhibit weaker capabilities in responding to dangerous situations encountered during home visits. The current nurse–patient relationship is a significant concern, and in the event of disputes, the personal safety of nurses is challenging to ensure [92]. Therefore, international scholars suggest the Preparedness, Awareness, Alertness, and Prevention (PAAP) strategy for implementation before and during home visits to protect home health care workers. Institutions can also enhance safety by offering training based on the PAAP strategy [93,94]. Moreover, to ensure nurses' safety, implement mobile location tracking, platform monitoring, one-touch alarm devices, and video recording for process traceability. Both nurses and patients should have insurance [95]. Furthermore, male nurses have notable safety benefits over female

nurses when caring for overweight or paralyzed patients due to their physical strength and emergency handling skills, like lifting patients, performing Cardiopulmonary Resuscitation (CPR), and managing fractures. Thus, they may be well-suited for primary roles in home visits in the self-employed nursing sector in the future.

This study shows that self-employed nurses who are outside the traditional hospital system can directly charge patients for nursing services. Self-employed nurses manifest the economic value of nursing services through direct fee-for-service arrangements and achieve higher earnings than their hospital-employed counterparts. However, the lack of a unified pricing standard presents significant challenges in fee-setting. Self-employed nurses were concerned with setting too high or too low fees. When determining their fees, they must consider personal value, market acceptance, and additional costs such as taxes and insurance [96]. Currently, in China, the pricing of "online nurses," closely related to self-employed nursing, lacks a unified standard on internet platforms. The cost of home visits is much greater than that of hospital-based services, and there is a significant variation in fees across different platforms. Moreover, medical insurance does not cover these service fees, making it difficult for families with limited financial resources to afford the high costs. Therefore, service fees must be reasonably regulated [92]. Home visit fees should be capped on the basis of the service prices of medical institutions [97]. In 2010, Australia reformed legislation to boost community and primary care nurse practitioners, allowing self-employed nurses to offer subsidized care through government-funded programs [56]. China could adopt similar insurance benefits, like partial reimbursements or fee reductions.

4.3. Value of self-employed nurses

This study revealed that self-employed nurses benefit patients, health care professionals, and the health care system. In primary health care settings, self-employed nurses are capable of addressing most health issues typically managed by primary care physicians, thus decreasing the time and expenses linked to obtaining medical treatment for patients in low-income, remote regions and for those not covered by the national health care system. However, rural areas often face a shortage of health care personnel. International studies have shown that policy support, such as financial subsidies, improved working conditions, and career development opportunities, can effectively attract and retain nursing staff in remote regions [98]. Therefore, comprehensive social security benefits, including pension insurance, medical insurance, and housing subsidies, should be provided for rural nurses. A rational salary standard should be established to ensure that rural nurses receive wages comparable to those of their urban counterparts, and a remote area allowance should be set up to compensate for the inconveniences of living conditions in rural areas [99].

Self-employed nurses provide health care professionals with innovative and autonomous nursing roles. Self-employed nurses transcend standardized institutional protocols by proactively designing service content and models based on patient needs and professional nursing values. They exercise flexibility in scheduling. However, there is currently a lack of training related to self-employed nurses. Shirey [18] highlighted that the lack of business skills is a barrier for nurses considering self-employment. In China, nursing students typically do not receive training related to self-employment during their formal education. This educational gap may limit young nurses' opportunities to choose self-employment, resulting in a lack of essential knowledge and skills when starting their self-employed businesses [75,100]. Therefore, educational interventions such as themed activities and disseminating exemplary outcomes on campus can be implemented. Moreover, educational authorities must emphasize entrepreneurial education in nursing and assist universities and colleges in offering relevant courses, including innovation and entrepreneurship programs. In addition to general business topics, education on self-employment for nurses should consider local laws and regulations and help them learn how to plan, organize, finance, and operate their businesses [101].

4.4. Registration and legalization

An increasing number of countries have formally integrated self-employed nurses into their regulatory frameworks, establishing mature pathways for registration and legalization. These developments offer valuable reference points for other nations formulating policies. In Canada, legislatively mandated provincial regulatory bodies safeguard public safety and welfare by promoting competent practice, preventing harmful practice, and investigating allegations of misconduct or malpractice [15]. However, cumbersome regulatory procedures may present challenges for self-employed practitioners. These nurses report negative experiences with restrictive processes, delayed approvals, excessive monitoring, inconsistent application, and lack of transparency in regulatory oversight, fostering distrust, fear, and helplessness toward regulatory authorities [8,15,20]. In countries like China, the criteria for admitting self-employed nurses and the processes for reviewing and certifying their qualifications are inconsistent and unclear. Some private platforms only require a nursing qualification or practice certificate, lacking specific demands for experience, specialized skills, professional titles, or practice records [69,102]. Therefore, there is an urgent need to enact relevant national policies to unify the admission standards for self-employed nurses. A thorough evaluation process for nurses' qualifications and continuous oversight is crucial for healthcare quality and safety.

4.5. Policy recommendations

To promote the healthy development of the profession of self-employed nurses, the policy system must be improved from multiple dimensions. First, the access standards for individual professional nurses should be unified. Drawing on international experience, qualifications and abilities should be strictly reviewed, and a standardized certification system should be established. Moreover, the clinical experience requirements for self-employed nurses in specialized fields can be shortened to facilitate workforce expansion.

Second, a risk prevention and emergency management mechanism should be established. Targeted training should be conducted, and the legal responsibilities of self-employed nurses should be clarified. The information-sharing and cooperation model should be improved, and technical means should be employed to ensure the personal safety of nurses. The advantages of male nurses in home visits should also be explored.

Third, service fee pricing should be standardized. A cap on home visit fees should be set, and insurance and welfare subsidy policies should be introduced to increase service accessibility. For rural areas, incentive policies such as financial subsidies, improved treatment, and enhanced training should be implemented to attract and retain nursing talent.

Finally, in the field of education, nursing entrepreneurship education should be emphasized. Related courses should be offered, and educational intervention activities should be performed to enhance nursing students' understanding and practical

abilities in self-employment.

4.6. Limitations

Although this study systematically explores the role and value of self-employed nurses, several limitations remain. Given that the identified literature predominantly represents high-income countries, with low-income and middle-income countries being underrepresented, this reflects geographical and practice-context biases. Another limitation of this study is related to policy. Some countries may be unable to implement self-employed nursing due to a lack of policies, and some of the proposed policies may not apply to all countries.

5. Conclusions

This study conducted a scoping review of the role and value of self-employed nurses. Self-employed nurses must hold a bachelor's degree in nursing and pass relevant examinations; specialist nurses face even higher demands and must possess diverse skills. These nurses can provide home visits and personalized patient care through clinics or home visits. Their practice offers health care workers a new career pathway with greater flexibility and autonomy, reduces health care costs, and fosters interdisciplinary collaboration. However, they face several challenges, including inconsistent regulations, risks, a lack of medical insurance policies, insufficient staffing in remote areas, and inadequate educational support. Standardizing entry requirements, strengthening certification, enhancing safety, developing medical insurance policies, attracting talent, and reinforcing education are key strategies to enable them to contribute effectively to the health care system.

Data availability statement

Data sharing is not applicable to this article, as no datasets were generated or analyzed during the current study.

CRediT authorship contribution statement

Nianqi Cui: Conceptualization, Methodology, Formal analysis, Validation, Resources, Funding acquisition, Writing - original draft, Writing - review & editing. **Wei Yang:** Conceptualization, Methodology, Formal analysis, Data curation, Formal analysis, Writing - original draft, Revising, Writing - review & editing. **Yulin Lu:** Conceptualization, Methodology, Supervision, Project administration, Resources, Funding acquisition, Writing - review & editing. **Jingfen Jin:** Conceptualization, Methodology, Validation, Supervision, Project administration, Resources, Writing - review & editing.

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Declaration of competing interest

The authors of this study have declared no conflict of interest.

Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijnss.2025.08.005.

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